**ADSP 31014 Statistical Models for Data Science**
**Homework 5**

**(10 points) Question 1a**
**Initial Model:** Let's consider our `chicago_taxi_trip_sample.csv` data set. Fit the following linear regression model with `trip_seconds` as the response variable and some quantitative explanatory variables:

```
lm(trip_seconds ~ extras + fare + tips + tips_percent + tolls
                  + trip_miles + trip_total, ...)
```

- What are the AIC and adjusted $R^2$ values of the fitted model?
- Based on the t-test, which explanatory variable is the least useful? What is its p-value?

**(10 points) Question 1b**
**Backward Elimination:** Continuing from Question 1a, perform backward variable elimination using AIC as the criterion until convergence.

- What is the reduced model formula? What are the AIC and adjusted $R^2$ values of the reduced model?
- Based on the t-test, which one of the remaining explanatory variables is the least useful? What is its p-value? Based on this p-value, should we further eliminate this explanatory variable?

**(15 points) Question 1c**
**Forward Selection:** Let's consider the null base model

```
lm(trip_seconds ~ 1, ...)
```

- Perform forward variable selection using AIC as the criterion starting from this base model up to the scope of the full model in Question 1a.
- What is the model formula of the selected model? What are the AIC and adjusted $R^2$ values of the selected model?
- Is this selected model same as the reduced model in Question 1b?

**(15 points) Question 1d**
**Borderline Variables:** Let's investigate the disagreement between forward selection in Question 1c and backward elimination in Question 1b.

- Starting from the reduced model in Question 1b, go ahead and further eliminate the least useful remaining explanatory variable identified in Question 1b.
- What are the AIC and adjusted $R^2$ values of this further reduced model?
- Interestingly, now another explanatory variable in this further reduced model suddenly becomes not very useful. Which explanatory variable is it? What is its p-value based on t-test?

**(5 points) Question 1e**
**Best Model:** Among all the fitted models in Question 1a-1d, which one is the best based on AIC? Based on adjusted $R^2$?

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**(10 points) Question 2a**
**"I'm Lazy" Model:** It is usually not a good idea to blindly include all possible interaction terms into a model because the number of parameters could explode very quickly. Nonetheless, let's fit the following linear regression model with trip_seconds as the response variable and all interactions terms among fare and some categorical explanatory variables:

```
lm(trip_seconds ~ fare * factor(day_of_week) * hour_type
                  * pickup_area * dropoff_area * payment_type, ...)
```

- How many $\beta$'s parameters does this model have?
- What are the AIC and adjusted $R^2$ values of the fitted model?

**(10 points) Question 2b**
**Hope or Hopeless:** One might hope that backward elimination could always help us simplify an overly-complicated model.

- Continuing from Question 2a, perform backward variable elimination using F-test as the criterion until convergence.
- Which variables or terms are eliminated? (Hint: How many $\beta$'s parameters does this model have?)

**(10 points) Question 3a**
**Large Data Set vs Tiny Effect:** If we have a lot of data, then we can detect even a tiny effect i.e. a tiny deviation from the null hypothesis $H_0$. Let's do a simulation to understand this.

Let $X_1, \ldots, X_n \sim \text{Bern}(0.5 + 10^{-5})$ be an i.i.d. sample of $n$ observed data points. Think of these as coin tosses from a slightly unfair coin. Let's consider the two-sided binomial proportion test for

$$H_0: p = 0.5 \quad \text{vs} \quad H_1: p \neq 0.5$$

If $n$ is sufficiently large, then the computed p-value should be small and we would reject $H_0$ correctly. The following R code simulation computes the p-value when $n = 10^5$. (Technically, we repeat the procedure 10 times and take the maximum of those 10 p-values in order to reduce the luck factor of encountering one small p-value just by chance.)

```
num.of.repeat <- 10
p.values <- numeric(num.of.repeat)
for (i in seq_len(num.of.repeat)) {
  # Step 1: Simulate observed data x_1,...,x_n ~ Bern(p1) under H1
  n <- 1e5
  p1 <- 0.5 + 1e-5
  x <- rbinom(n = 1, size = n, prob = p1)
  # Step 2: Compute p-value of two-sided binomial proportion test H0: p = 0.5
  p.values[i] <- prop.test(...)$p.value
}

max(p.values)
```

Complete the R code above and run it once. What is the resulting value of `max(p.values)`?

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**(5 points) Question 3b**
**Small P-Value:** In Question 3a scenario, how large must $n$ be so that `max(p.values)` < 0.01? (Hint: Try $n = 10^k$ for various values of $k$.)

In conclusion, if you toss a coin a sufficiently large number of times, then you could detect even a tiny manufacturing imperfection of an unfair coin. In the context of linear regression, if we have a lot of data, then F-test or t-test (such as Question 2b) tends to reject the null hypothesis $H_0$ (that an explanatory variable is not useful). In other words, this kind of statistical model selection technique tends to yield a larger model with more explanatory variables.

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